



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

We need to know how to track defects.



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Ask your partner:

- How do you think that collecting data might help a project to succeed?
- What do you think is the most important kind of data you need to collect when tracking defects?

Can Do Statements:

- I can talk about the importance of collecting data when defect tracking.
- I can recall experiences that show the importance of collecting data when defect tracking.
- I can discuss the positive outcomes of refining the process of collecting data.





Let's Explore!

There are some misconceptions* about defect tracking

* We use the word "*misconception*" when we want to talk about a mistaken idea about something.



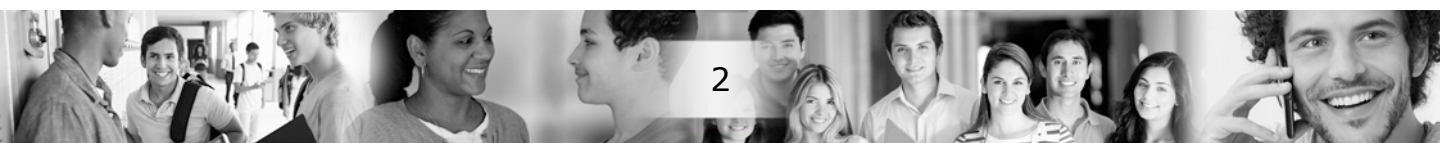
Instructions: Read the text below.

Collecting data for defect tracking is a pivotal process for our company, and we wouldn't have been able to fix so many defects if it weren't for this process. However, our organization was having trouble with the excessive amount of data it was dealing with. We shouldn't have thought that by entering as much data as possible, we were going to improve the defect tracking process. This process could have been less tedious, but we weren't selective with the data we collected.

Although we were reluctant at the beginning, since we thought that any change could have affected our CMMI level 4, we decided to evaluate the company's processes performance to get an overview of the project. To do so, we built a flow diagram of the processes the team was carrying out, but we noticed that not all of the processes that were stated by the CMMI were actually being used by the team members.

We realized that we were carrying out the defect tracking process in a counterproductive way, as the team were entering unnecessary data; thus, they were losing time when doing so because they were not going back to analyze and use all of the gathered information. Therefore, we had to streamline the process of defect tracking so that the collected data was relevant for the team. This helped the team collect the defect tracking data without jeopardizing the CMMI level. All in all, we wouldn't have strengthened the effectiveness of the team if we hadn't come up with a better way of gathering data.

Defect tracking allowed us to collect significant information to improve our services for better, but we wouldn't have fixed so many defects if we hadn't undergone this process. Unfortunately, we could have collected more relevant data and become more efficient in fixing defects if we had refined our defect tracking process before.





Let's Practice!

This could have been so different

Instructions: Find the missing words for each sentence.

1. Some bad decisions made by the CEO could have _____ the entire project's maturity level.
2. If they wanted to improve the maturity level of the project, some processes should have been _____.
3. The software team shouldn't have been so _____ to adopt the recommended model practices.
4. A CMMI _____ level 4 would have been achieved, two weeks ago, if we had followed the CEO recommendations.
5. Data collection could have been a _____ and repetitive process, but the new method made it much faster.

I	J	E	O	P	A	R	D	I	Z	E	D	R	H	Z
Z	S	P	P	D	A	S	P	Q	G	W	D	Y	F	H
B	G	M	Y	Z	P	T	A	B	U	T	C	I	X	M
V	G	S	H	V	Y	R	R	I	H	E	X	G	M	N
B	A	Z	V	L	N	E	P	G	C	D	M	W	A	C
W	I	T	Q	I	G	A	M	P	E	I	O	O	T	I
J	B	O	M	C	L	M	V	P	F	O	Z	X	U	J
R	B	M	N	I	I	L	Y	Y	B	U	R	X	R	R
T	L	E	D	Y	F	I	K	X	Z	S	G	Z	I	E
N	D	H	R	O	A	N	U	L	H	S	K	F	T	B
C	P	J	G	J	D	E	B	Q	B	A	D	Z	Y	Y
A	Q	N	D	E	Z	D	M	O	I	W	H	C	I	A
Y	X	S	H	Y	A	G	U	N	H	S	I	Z	B	M
S	R	E	L	U	C	T	A	N	T	F	I	U	C	V
U	F	O	S	C	Z	R	V	L	D	E	A	L	D	V



Let's Practice!

What would have happened if...

Instructions: Proofread the sentences. Correct the mistakes.

1. They thought that any change would having affected their CMMI level 4, as seen in the flow chart.
2. We can have collected more relevant data if we had refined our defect tracking process before.
3. The team shouldn't have entering so much data; the defect tracking process is going to take weeks.
4. We wouldn't have being able to fix so many code defects if it weren't for the new recommended practices.
5. We can't have strengthened the effectiveness of the team if we hadn't hired expert coders.



Extension Activity

What is the importance of collecting data when implementing defect tracking?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information that states how collecting data in defect tracking has been useful for a project.
- Write a short text (200 words) where you explain how it helped the project to succeed.
- Share the text with your partners.

Self-Assessment

Yes

No

- I investigate about collecting data in defect tracking.
- I write about collecting data when defect tracking.
- I talk about collecting data in defect tracking.

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Target Words

- **reluctant** (adjective): something that is done unwillingly.
e.g. Although we were reluctant at the beginning, since we thought that any change could have affected our CMMI level 4, we decided to evaluate the company's processes performance to get an overview of the project.
- **flow diagram** (noun): a diagram that shows the dynamic relationships of a system.
e.g. To do so, we built a flow diagram of the processes the team was carrying out.
- **come up with** (phrasal verb): to think of an idea or solution for something.
e.g. We wouldn't have strengthened the effectiveness of the team if we hadn't come up with a better way of gathering data.
- **streamline** (verb): to change something for making it more efficient.
e.g. Therefore, we had to streamline the process of defect tracking so that the collected data was relevant for the team.
- **jeopardize** (verb): to hazard, to endanger.
e.g. This helped the team collect the defect tracking data without jeopardizing the CMMI level.
- **counterproductive** (adjective): something that is not productive or efficient.
e.g. We realized that we were carrying out the defect tracking process in a counterproductive way.



Learning Points

Modal Verbs to Talk Hypothetically about the Past (Would have, Should have, and Could have)

We can use **modal verbs** like **could**, **should** or **would** when we want to talk about a possible event that could have happened in the past but did not.

The structure of the modal verbs used to talk hypothetically about the past is the following:

Affirmative: **Modal verb + have + past participle of the main verb.**

Negative: **Modal verb + not + have + past participle of the main verb.**

Example:

We **should have collected** only the necessary data.

We **should not have collected** so much data.

Let's take a look at the meaning of the different modal verbs.

Could

Affirmative: It is used to talk about something that was likely to happen but did not.

Example:

The **defects could have been** better addressed with the appropriate data.

Negative: It is used to talk about something that was not possible to happen in the past.

Example:

The **defects could not have been detected**, even with the appropriate data.

Should

Affirmative: It is used to express that the ideal was for something to happen, but it did not.

Example:

We **should have tracked** the defects from the beginning.

Negative: It is used to express that the ideal was for something not to happen.

Example:

We **should not have tracked** the defects from the beginning.

Would

Affirmative: It is either used to express something you wanted to happen but did not or as part of the main clause in the third conditional.

Example:

Too bad we did not know about CMMI before, **we would have improved** the performance of the process areas.

If we had taken into consideration the importance of the processes needed to develop software, **we would have been** much more effective and obtained a higher quality product.

Negative: It is used to express a choice someone was unlikely to have taken.

Example:

We **would not have improved** the performance of the process areas.





Learning Points

Examples

- *Collecting data for defect tracking is a pivotal process for our company, and we wouldn't have been able to fix so many defects if it weren't for this process.*
- *We shouldn't have thought that by entering as much data as possible, we were going to improve the defect tracking process.*
- *This process could have been less tedious, but we weren't selective with the data we collected.*

What is CMMI?

CMMI stands for Capability Maturity Model Integration. It is a process model that describes the characteristics of effective processes.

The processes that an organization follows for the development of software are very important. A lot of the general performance of the company is going to be determined by the effectiveness of the processes it undertakes.

Companies apply CMMI principles in the planning and execution of their processes, as a collection of best practices that will improve the effectiveness of their activities, identify and achieve measurable business goals, and build better products.

An organization can be categorized into different **maturity levels**. These levels are achieved once the organization is able to fulfill a set of **process areas**. Process areas describe the aspects of product development that need to be covered by the organization, and they are meant to adapt to the behavior of the company. Let's take a look at some process areas in the maturity levels:

Maturity Level 1: Initial

This level is for organizations that do not have any maturity level.

Maturity Level 2: Managed

Measurement and Analysis
Project Monitoring and Control
Project planning
Requirements Management
Etc.

Maturity Level 3: Defined

Decision Analysis and Resolution
Integrated Project Management
Requirements Development
Risk Management
Etc.

Maturity Level 4: Quantitatively Managed

Organizational Process Performance
Organizational Project Management

Maturity Level 5: Optimizing

Causal Analysis and Resolution
Organizational Performance Management

